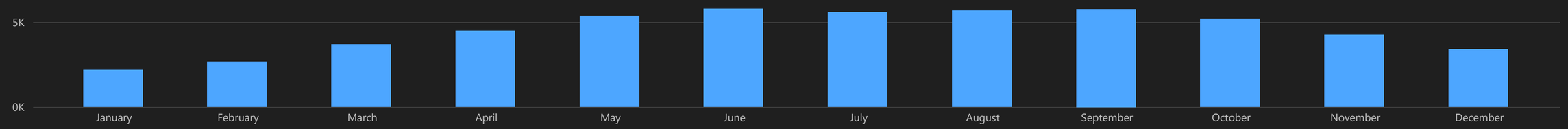
BIKE-SHARING ANALYTICS

Rental demand analysis | 2011-12

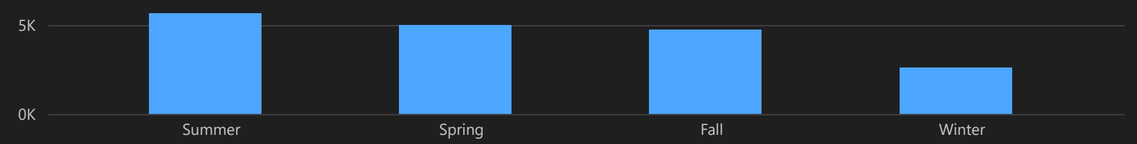


Total Rentals	Average Daily Rentals	Casual Rentals	Registered Rentals
3M	4.50K	620K	3M

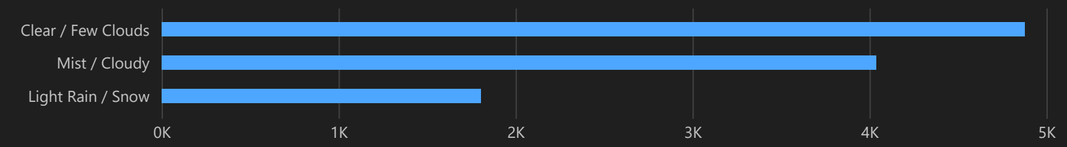
Average Daily Rentals
by Month Name



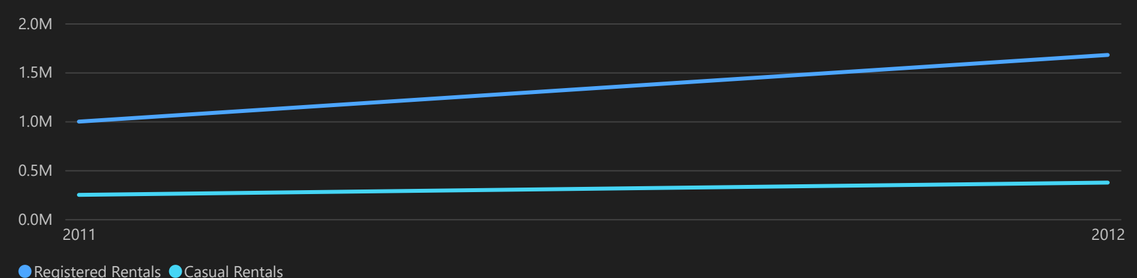
Average Daily Rentals by Season
Average Daily Rentals by Season2



Average Daily Rentals
by WeatherCondition



Casual Rentals vs Registered Rentals by Year
Registered Rentals and Casual Rentals by Year



WeatherCondition
Select WeatherCondition

Year
Select Year

Clear all slicers

Day Type
Select Day Type

Season2
Select Season2

BIKE-SHARING: EXECUTIVE NARRATIVE

Project overview, analytical approach and objectives

EXECUTIVE NARRATIVE

A concise overview of the dataset, dashboard purpose, and analytical approach.

The Bike-Sharing Analytics project analyses daily bicycle rental data from 2011 and 2012 to understand demand patterns and user behaviour. The dataset includes rental counts, seasonal information, weather conditions, calendar variables, and environmental measurements.

The dashboard presents key performance indicators, including total rentals, average daily rentals, casual rentals, and registered rentals. Interactive filters allow users to explore rental demand by year, season, weather condition, and day type.

The analysis examines monthly rental trends, seasonal demand variations, weather-related differences, and the distribution of rentals between casual and registered users. These comparisons help identify periods of higher and lower demand.

The primary objective is to support data-driven decisions about bicycle availability, resource allocation, and operational planning. The findings describe patterns within the observed dataset and should not be interpreted as proof of causation.

Dataset scope: The analysis uses daily observations from 2011–2012. The findings may not represent current demand or other bike-sharing systems.

KEY INSIGHTS

KEY INSIGHTS — WHAT THE DATA REVEALS

01. Rental Demand Varies Across Months
The monthly rental analysis shows clear variations in average daily rentals throughout the year. Demand rises during the middle months and declines toward the end of the year. This pattern can help identify periods when bicycle availability may need to increase.

02. Seasonal Demand Shows Clear Differences
The seasonal comparison indicates higher average daily rentals during summer and lower rentals during winter. Seasonal patterns can support demand forecasting and operational planning.

03. Weather Conditions Are Associated With Rental Demand
The dashboard shows higher average rentals under clear or partly cloudy conditions than under light rain or snow conditions. Weather information can help operators anticipate changes in demand.

04. Registered Users Account for Most Rentals
Registered rentals substantially exceed casual rentals in the dataset. Understanding the behaviour of these two user groups can help improve customer retention and service planning.

05. Rental Demand Increased Between 2011 and 2012
The yearly comparison shows an increase in both casual and registered rentals from 2011 to 2012. This indicates growth in recorded rental activity during the period studied.

Interpretation note: These findings describe patterns in the observed dataset. They do not establish that season or weather alone caused changes in rental demand.

RECOMMENDED SOLUTIONS

RECOMMENDATIONS — TURNING INSIGHTS INTO ACTION

01. Improve Bicycle Availability During High-Demand Periods

Use monthly and seasonal rental patterns to plan bicycle availability. Consider allocating additional bicycles during periods of higher demand and reviewing availability during quieter months.

02. Plan Maintenance During Lower-Demand Periods

Use historical rental trends to identify potentially quieter periods for routine maintenance and repairs. Schedule maintenance carefully to avoid reducing availability when demand is high.

03. Incorporate Weather Into Operational Planning

Monitor weather forecasts alongside historical rental patterns. Prepare for potential changes in demand during adverse weather conditions and adjust staffing and bicycle availability where appropriate.

04. Understand Casual and Registered Users

Analyse rental behaviour across casual and registered users. Use these findings to improve service planning, encourage repeat usage, and develop appropriate customer engagement strategies.

05. Monitor Demand Trends Regularly

Update the analysis as new rental data becomes available. Compare monthly, seasonal, and yearly trends to identify changes in demand and support evidence-based operational decisions.

06. Use Dashboard Filters for Detailed Analysis

Use the year, season, weather, and day-type slicers to investigate specific conditions. Review filtered results before making operational decisions, rather than relying only on overall averages.

Expected Benefits

These recommendations can support better resource allocation, maintenance planning, and demand monitoring. Their effectiveness should be assessed using future rental data rather than assumed in advance.